

or partial migration), the processes involved also differ. Broadly speaking, the various cloud migration types are:

1. **Lift and shift** – Lift and shift is the process of moving the software from on-premise to the cloud without making any changes to the application. This is the fastest method of migrating applications to the cloud, and causes least disruption. For a lift and shift migration, only the infrastructure and security teams need to be involved. The upfront costs are minimal. However, since it makes no changes to the application architecture or code, the performance is not optimised by exploiting the power of the cloud. Hence, it usually doesn't lead to long-term cost savings. Lift and shift is best suited for applications with regular peak schedules and predictable market changes, such as applications of food delivery companies and tax applications.
2. **Shift to SaaS** – Shift to SaaS involves outsourcing one or more applications to a specialised cloud service provider. By this method, companies can focus on their core applications and offload the less business-critical applications to SaaS providers. Its main advantage is that the company can remain competitive by streamlining its core business. However, although you can personalise your outsourced application, you can lose the support model to the SaaS company; hence, it is advisable to outsource only applications used for routine functions that are not customer-facing. Outsourcing the company email to Office 365 is an example of this type of migration.
3. **Application refactoring or app modernisation** – Refactoring requires some development effort. It is a method of prioritising and first

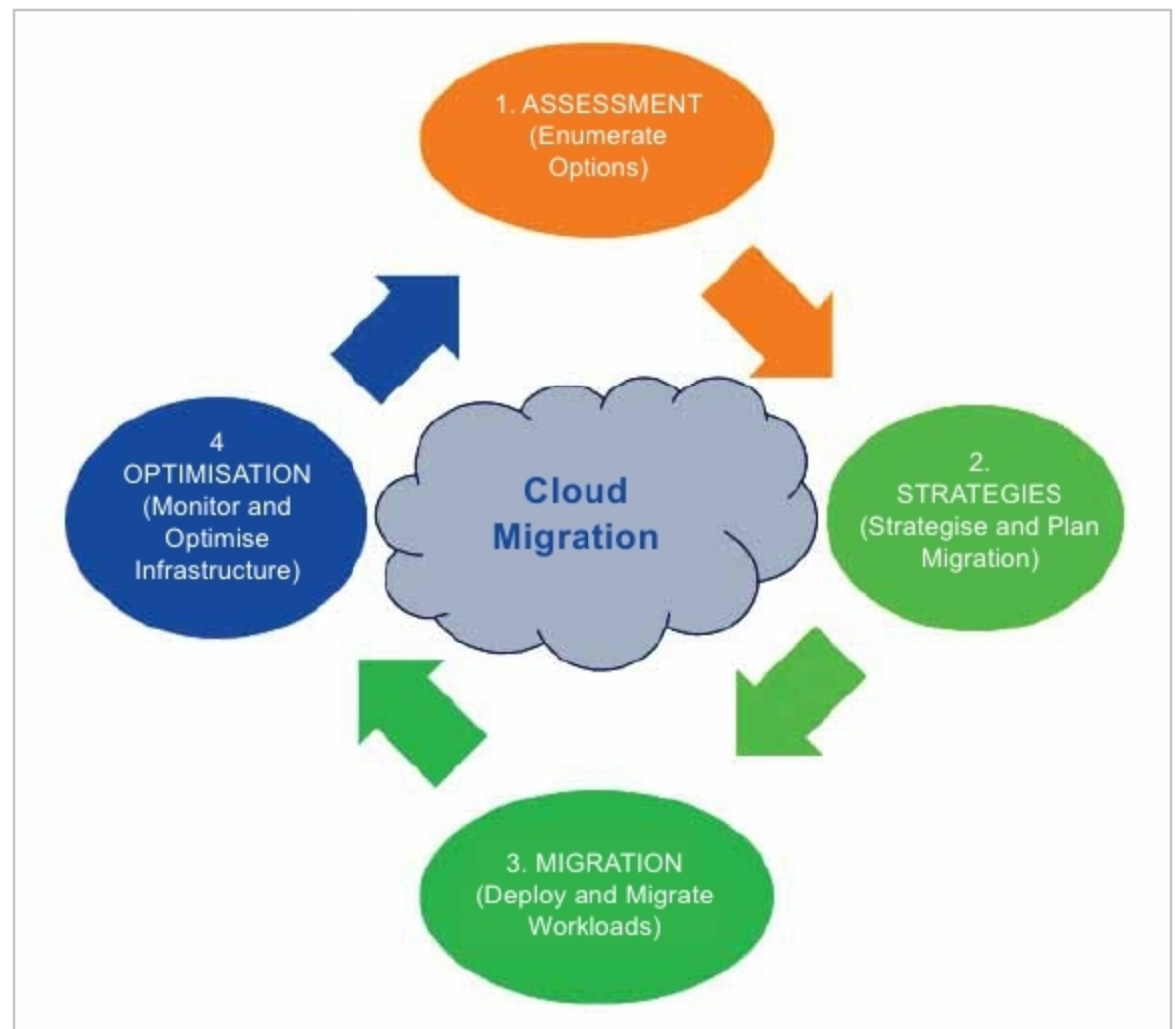


Figure 1: Cloud migration stages

moving the less critical applications that could benefit from the cloud's agility. It is a low-risk process, since the legacy applications can work in parallel with the new applications developed in the cloud.

While using cloud-native services helps to cut costs, refactoring also saves the time taken to migrate.

4. **Re-platforming** – This involves a code-rewrite and re-architecture of the complete application to make it cloud-native. It is the most resource-intensive migration process and requires a lot of planning. It is also the costliest type of cloud migration. However, by re-platforming the application, the organisation is able to get the complete benefit of the cloud, which in the long run proves very beneficial to the business.

Security considerations of cloud migration strategy

Before planning cloud migration, organisations must have their security

control policies and procedures in place. The security strategy should ensure unified policy enforcements and procedures for secure access to information across cloud environments. These policies should govern access and control regardless of whether the security gateways and services being used reside on-premise or in the cloud.

Another important aspect of security is to determine the regulatory and compliance requirements when data gets transmitted between various cloud and physical network environments. Organisations need to plan ahead to stay compliant and to secure the infrastructure. They also need to plan the right security solutions when they migrate their applications to the cloud.

Implementing several security tools will have to be considered when the area of attack increases post-migration. Apart from firewalls, solutions such as Web application firewalls, intrusion prevention and detection systems, privileged identity management, dual